

Palm VII™ SDK
INetLib

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Message

- Wireless capability can easily be incorporated into your existing or new C/C++ Palm™ applications using the INetLib API
- INetLib is necessary for using the Palm VII™ wireless radio

Agenda

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- The Palm VII™ SDK
- INetLib
- Development Resources
- Conclusion
- Q&A


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The Palm VII™ SDK

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- What is it?
  - The software development kit for the Palm OS® 3.2
  - The same APIs as the Palm OS 3.0 SDK, but with additional APIs for the wireless internet feature set
    - INetLib
- Where is it?
  - Bundled with Metrowerks CodeWarrior R6
  - Development Resources Seeding Program of www.palm.com/devzone web site
- How is it used?
  - Include the SDK directories in your compiler path


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INetLib

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What Is INetLib?

- An API designed to simplify Internet access
  - Wireless or wireline
- Protocols
  - HTTP, HTTPS, FILE, MailTo, Palm, and Palmcall
- Built on top of the NetLib API
  - Similar to Berkley Sockets API

INetLib

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Why Use INetLib?

- Easy wireless Internet access
  - Need INetLib to use Palm VII™ wireless modem
    - Code for communicating with wireless modem is part of INetLib
- Modification of hand-held databases with data from Internet servers
- Whatever you can dream up

INetLib

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Stability

- Major changes not anticipated
- Minor changes expected in the future
  - Watch for notices in the Development Resources Seeding Program
  - Warning in the reference doc is more a reflection of WebLib

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Header Files

- INetMgr.h
  - Header file for INetLib API declarations and constants
- CTP.h
  - Constants for clipping proxy server instructions
    - e.g., the conversion algorithm to use

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INetLib

In the Beginning…

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INetLib—In the Beginning…

Prerequisites

- INetLib must be available

```
// check if INetLib is installed
err = FtrGet(inetLibFtrCreator,
             netFtrNumVersion,
             &value);

// find the reference number
if(!err && value)
  err = SysLibFind("INet.lib", &libRefNum);
if(!err)              // then you're ready
```

INetLib—In the Beginning…

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To start

- INetLibOpen
  - Opens the NetLib and INetLib
    - Extra 33 Kb of RAM
    - More when using HTTPS and/or cache
  - Provides an inet handle
    - Unique to your app
  - Retrieves default settings
    - Network or Wireless Preferences panel
  - Allows you to specify your own configuration
    - Default wireless and wireline configurations available, e.g.,

      `inetCfgNameDefWireless`

      `inetCfgNameDefWireline`

INetLib—In the Beginning…

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Example code

```
INetConfigNameType cfgName =
                        { inetCfgNameCTPDefault };
                        // web clipping config

err = INetLibConfigIndexFromName(libRefnum,
                                 &cfgName,
                                 &cfgIndex);
err = INetLibOpen(
                libRefnum,
                cfgIndex,
                0,      // flags not used
                NULL,   // cache to use
                0,      // max cache size
                &inetH);
```

INetLib—In the Beginning…

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Customizing

- INetLibSettingSet
  - Affects only the inet handle for the calling app
  - Changes in affect until INetLibClose
  - Used for changing
    - The size of the cache
    - The configuration to use (index value)
    - Maximum response size (1024 bytes)
    - Conversion algorithm (ctpConvNone)
    - Display width for content (160 pixels)
    - If zipcode or deviceid data can be sent

INetLib—In the Beginning…

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Example code

```
UInt32 convType = ctpConvNone;

// set conversion algorithm for raw data
err = INetLibSettingSet(
                libRefNum,
                inetH,
                inetSettingConvAlgorithm,
                (BytePtr) &convType,
                sizeof(UInt32)
                        );
```

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The Two Roads

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INetLib—The Two Roads

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The High Road

- INetLibURLOpen
  - Basics
    - Pass a URL
      - Protocol extracted from given URL
      - Handles all of the supported protocols (schemes)
    - Returns a socket handle to read
    - INetLibGetEvent in event loop instead of EvtGetEvent
  - Equivalent to

```
INetLibSockOpen
INetLibSockConnect
… routines for handling protocols
```

INetLib—The Two Roads

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The High Road

- INetLibURLOpen
  - Other parameters
    - Connection timeout, e.g.,
      - 1 minute = 60 * SysTicksPerSecond();
    - Flags, e.g.,
      - inetOpenURLFlagLookInCache &
      - inetOpenURLFlagKeepInCache

INetLib—The Two Roads

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The High Road

- Example code

```
err = INetLibURLOpen(
                    libRefnum,
                    inetH,
                    &url,
                    &url,  // index using same url
                    &sockH,
                    timeout,
                    inetOpenURLFlagLookInCache &
                    inetOpenURLFlagKeepInCache
                        // url in cache ...
                );
```

INetLib—The Two Roads

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The Low Road: HTTP or HTTPS

- Only for HTTP or HTTPS
  - You could write parsing code, etc. to handle FILE, Palm, PalmCall, etc.
    - Suggest using INetLibOpenURL instead
- INetLibSockOpen
  - Opens a socket for the given protocol (scheme)
    - Scheme types
      - inetSchemeHTTP and inetSchemeHTTPS
  - Sets the socket handle argument

INetLib—The Two Roads

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The Low Road: HTTP or HTTPS

- INetLibSockSettingSet
  - Customize the socket
    - inetSockSettingFlags
      - inetOpenURLFlagLookInCache
      - inetOpenURLFlagKeepInCache
    - inetSockSettingReadTimeout
      - Timeout value for reading from a socket
    - inetSockSettingTitle
      - Title of cached item
    - inetSockSettingIndexURL
      - Used to group responses together

        e.g., associate history list clippings with a WCA

INetLib—The Two Roads

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The Low Road: HTTP or HTTPS

- INetLibSockConnect
  - Initiates the connection attempt
  - May return before connection established
    - No blocking
    - Monitor inetSockStatusChangeEvent
  - Arguments
    - Pointer to host name string
      - Dotted decimal OK
    - Port number
    - Timeout value

INetLib—The Two Roads

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The Low Road: HTTP or HTTPS

- INetLibSockHTTPReqCreate
  - Forms an HTTP or HTTPS request
    - Don’t use for other protocols
  - Pass the method
    - e.g., GET or POST
  - Pass a pointer to the URL

INetLib—The Two Roads

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The Low Road: HTTP or HTTPS

- INetLibSockHTTPAttrSet
  - inetHTTPAttrIncHTTP
    - When set to true and ctpConvNone, includes HTTP headers in raw data

INetLib—The Two Roads

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The Low Road: HTTP or HTTPS

- INetLibSockHTTPReqSend
  - Sends the HTTP or HTTPS request
    - Checks the cache first if **`inetLibOpenURLFlagLookInCache`** previously set
  - Pass pointer to content data
    - For POST method
  - Pass timeout

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Wrap It Up

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INetLib—Wrap It Up

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Monitoring

- INetLibGetEvent
  - Replaces EvtGetEvent
  - Notifies of status changes in
    - Internet sockets
    - UI
  - inetSockReadyEvent
  - inetSockStatusChangeEvent, e.g.,
    - Connecting, sending, waiting, receiving

INetLib—Wrap It Up

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Retrieving Data

- INetLibSockRead
  - Reads data from a socket
  - Blocks until at least 1 byte read
  - Loop
    - Until 0 bytes read
      - Assuming inetSockReadyEvent
  - Arguments
    - Pointer to buffer to write the data
    - Number of bytes to read
    - Pointer to actual number of bytes read
    - Timeout
      - Useful to exit blocking state

INetLib—Wrap It Up

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Closing Down

- INetLibSockClose
  - Closes the socket
  - Frees memory
- INetLibClose
  - Closes the inet handle
    - Also closes any open sockets associated with the app
  - Frees memory
  - Open count kept to prevent closure if other apps using INetLib (NetLib function)
  - Timer started if open count decrements to 0 (NetLib function)

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Extras

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INetLib—Extras

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System Cache

- Least Recently Used (LRU)
- Same cache available for all apps
- Cached data stored in received format
- inetOpenURLFlagKeepInCache
  - Socket setting flag to store response in the cache
- inetOpenURLFlagLookInCache
  - To ensure checking of cache for URL
- Setting to use a custom cache database

INetLib—Extras

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System Cache utilities

- INetLibURLGetInfo
  - Checks if URL is cached
- INetLibCacheGetObject
  - Retrieves info about a cached item
  - Info stored in an **`INetCacheInfoType`** structure
- INetLibCacheList
  - Retrieves item from cache list by index

INetLib—Extras

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Compression

- Set by INetLibSettingSet
  - Using **`inetHTTPAttrConvAlgorithm`**
- Checked with INetLibSettingGet
  - **`inetSockSettingCompressionTypeID`**
- Types
  - **`inetCompressionTypeNone`**
    - Raw data through Palm.Net
  - **`inetCompressionTypeBitPacked`**
    - Standard web clipping format
  - **`inetCompressionTypeLZ77`**
    - Not yet supported

INetLib—Extras

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Some utilities

- INetLibSockStatus
  - Just opened, looking up host address, etc.
- INetLibURLsAdd
  - Appends URL fragments to base URL strings
  - Ensures resulting URL is in correct format

INetLib—Extras

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Some utilities

- INetLibURLCrack
  - Breaks a URL into various component parts
  - Stores parts in structure INetURLType
- INetLibURLGetInfo
  - Info on whether URL data was
    - Encrypted
    - Retrieved from the cache
    - Retrieved from the Internet


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Development Resources

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- Development Resources Seeding Program
  - Latest material
- The Palm VII™ SDK Docs
  - Reference
  - Programmer's Companion
- http://www.palm.com/devzone/assistance/
  - Mailing lists
  - Training info
  - Knowledge base
  - Newsgroup listings
  - Submit a technical question

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Conclusion

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Conclusion

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Summary

- The INetLib API provides easy Internet access
- Must use INetLib for internal Palm VII™ radio
  - Allows for wireline connections too
- Two means for making a wireless connection
  - Simple
    - INetLibOpenURL
  - More control
    - INetLibSockOpen
    - INetLibSockSettingSet
    - INetLibSockConnect
    - INetLibSockHTTPAttrCreate
    - INetLibSockHTTPAttrSet
    - INetLibSockHTTPAttrSend

Conclusion

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What next?

- Try it out in the labs at night
- Get the Palm VII™ SDK

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Q&A

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